K L UNIVERSITY

FRESHMAN ENGINEERING DEPARTMENT
A Project Based Lab Report on

Dynamic Transport Logistics and Route Optimizing Tool

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CERTIFICATE

This is to certify that the project based laboratory report entitled "Dynamic Transport Logistics and Route Optimization" submitted by **Eesh Nandan Pradhan,** bearing Regd. No. **2500031708** to the **Department of COMPUTER SICENCE ENGINEERING , KL University** in partial fulfillment of the requirements for the completion of a project based Laboratory in Python Programming (25SC2007)" course in I B Tech II Semester, is a Bonafede record of the work carried out by him/her under my supervision during the academic year 2025 – 2026.

PROJECT SUPERVISOR

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ABSTRACT

The Dynamic Transport Logistics & Route Optimization Tool (DTL-RO) is , a software application designed to optimize transportation routes by identifying the shortest and most efficient path between multiple locations. The system models real-world locations as nodes and routes as edges in a graph structure, enabling effective analysis of transportation networks.

The project utilizes Dijkstra's algorithm to compute the minimum distance between a selected source and destination. It takes user inputs such as locations and distances, processes them using graph-based techniques, and generates an optimized route along with the total distance.

This tool aims to reduce travel time, fuel consumption, and operational costs in logistics and transportation systems. It can be applied in real-world scenarios such as delivery services, supply chain management, and route planning for transport companies.

The modular design of the system improves code organization, readability, and scalability, making it suitable for further enhancements like real-time traffic integration and GPS-based navigation.

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INTRODUCTION

The **Dynamic Transport Logistics & Route Optimization Tool (DTL-RO)** is a software-based system developed to find the shortest and most efficient route between multiple locations.
It is mainly used in transportation, delivery services, logistics companies, and supply chain management.

In this project, locations such as cities, warehouses, or delivery points are represented as nodes, and the roads connecting them are represented as edges with distances.

The system uses Dijkstra's shortest path algorithm to determine the best possible route from a source to a destination.

The main purpose of this project is to minimize distance, travel time, fuel cost, and delivery delay.

AIM

The aim of this project is to design and develop a Dynamic Transport Logistics & Route Optimization Tool (DTL-RO) that efficiently identifies the shortest, safest, and most cost-effective route between multiple locations by using graph-based shortest path algorithms such as Dijkstra's algorithm.

The project aims to improve the overall efficiency of transportation and logistics operations by minimizing travel distance, reducing fuel consumption, saving time, lowering operational costs, and ensuring faster delivery of goods and services.

It is also intended to provide a scalable and modular platform that can be further enhanced with real-time traffic analysis, GPS integration, multiple vehicle routing, and smart decision-making capabilities for practical use in transport companies, delivery services, and supply chain management systems.

Advantages

:

- Finds shortest route quickly
- Reduces fuel consumption
- Saves travel time
- Improves delivery efficiency
- Easy to use
- Modular and scalable design
- Useful for logistics and transport companies

Disadvantages

- Works only with manual distance input
- Does not consider live traffic conditions
- No GPS or map integration
- Limited to distance-based optimization only
- Cannot handle road closures dynamically
- For very large networks, processing may take more time
-

Future Implementations

- Live traffic integration
- Google Maps / GPS support
- Real-time road condition updates
- Fuel cost estimation
- Travel time prediction
- Multiple vehicle optimization
- Delivery scheduling
- Weather-based route planning
- AI-based smart route prediction

SYSTEM REQUIREMENTS

➢ SOFTWARE REQUIREMENTS:

The major software requirements of the project
are as follows:Language: Turbo-C.
Operating system**:** Windows 10 or more.
Technical requirements: Monitor, CPU, Keyboard, Mouse, etc…

➢ HARDWARE REQUIREMENTS:

The hardware requirements that map towards the software

are as follows:RAM: 8 GB.

Processor: i5/i7.

FLOW CHART

START
Enter Number of Routes
Input Source, Destination & Distance
Store Data in Graph
Enter Start and End Location
Apply Dijkstra Algorithm
Find Shortest Route
Display Distance & Path
STOP

ALGORITHM

STEP 1: Start

STEP 2: Create empty graph

STEP 3: Read route details

STEP 4: Store routes in Graph

STEP 5: Input source and destination

STEP 6: Apply Dijkstra's algorithm

STEP 7: Find shortest route

STEP 8: Display optimized path and distance

9. Stop

IMPLEMENTATION

```
#, distance))
        self.graph[destination].append((source, distance))
graph_module.py

class Graph:
    def __init__(self):
        self.graph = {}

    def add_route(self, source, destination, distance):
        if source not in self.graph:
            self.graph[source] = []
        if destination not in self.graph:
            self.graph[destination] = []

        self.graph[source].append((destination
    def get_graph(self):
        return self.graph

#route_optimizer.py

import heapq

class RouteOptimizer:
    def __init__(self, graph):
        self.graph = graph

    def shortest_path(self, start, end):
        priority_queue = [(0, start, [])]
        visited = set()

        while priority_queue:
            cost, current, path = heapq.heappop(priority_queue)

            if current in visited:
                continue

            visited.add(current)
            path = path + [current]

            if current == end:
                return cost, path

            for neighbor, distance in self.graph[current]:
                if neighbor not in visited:
                    heapq.heappush(
                        priority_queue,
                        (cost + distance, neighbor, path)
                    )return float("inf"), []
```

```
#input_module.py

def get_route_input():
    routes = int(input("Enter number of routes: "))
    route_list = []

    for _ in range(routes):
        source = input("Enter source: ")
        destination = input("Enter destination: ")
        distance = int(input("Enter distance: "))
        route_list.append((source, destination, distance))

    start = input("Enter start location: ")
    end = input("Enter destination location: ")

    return route_list, start, end


#output_module.py

def display_output(distance, path):
    print("\n----- OPTIMIZED ROUTE------")
    print("Best Route:", " -> ".join(path))
    print("Minimum Distance:", distance, "km")

#main.py

from graph_module import Graph
from route_optimizer import RouteOptimizer
from input_module import get_route_input
from output_module import display_output

def main():
    graph_obj = Graph()

    route_list, start, end = get_route_input()

    for source, destination, distance in route_list:
        graph_obj.add_route(source, destination, distance)

    optimizer = RouteOptimizer(graph_obj.get_graph())

    distance, path = optimizer.shortest_path(start, end)

    display_output(distance, path)

if __name___== "__main__":
    main()
```

INTEGRATION AND SYSTEM TESTING OUTPUTS

Screen Shots:

```
Enter number of routes: 5
Enter source: Vijayawada
Enter destination: Mangalagiri
Enter distance: 12
Enter source: Mangalagiri
Enter destination: Amaravati
Enter distance: 25
Enter source: Vijayawada
Enter destination: Guntur
Enter distance: 32
Enter source: Guntur
Enter destination: Amaravati
Enter distance: 30
Enter source: Mangalagiri
Enter destination: Guntur
Enter distance: 18
Enter start location: Vijayawada
Enter destination location: Amaravati

----- OPTIMIZED ROUTE -----
Best Route: Vijayawada -> Mangalagiri -> Amaravati
Minimum Distance: 37 km

Process finished with exit code 0
```

CONCLUSION

The Dynamic Transport Logistics & Route Optimization Tool (DTL-RO) successfully demonstrates how graph-based algorithms can be used to solve real-world transportation and logistics problems. By applying Dijkstra's shortest path algorithm, the system efficiently finds the most optimal route between a source and destination.

The project helps in reducing travel distance, saving time, minimizing fuel consumption, and lowering transportation costs, making it highly useful for delivery services and logistics companies.

The modular design of the project improves readability, maintainability, and scalability, allowing future enhancements such as live traffic monitoring, GPS integration, and smart route prediction.

Overall, this project provides an effective and practical solution for route optimization in transport management systems and serves as a strong foundation for advanced logistics applications.